Transformer–Koopman: History-Conditioned Scheduled Prediction and Frozen-LTV Model Predictive Control

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***Abstract*—Koopman-inspired predictors provide a structured representation of nonlinear dynamics by evolving observables through finite-dimensional linear maps. A fixed lifted transition, however, can be restrictive when the measured state does not determine the relevant operating condition. This paper develops a history-conditioned scheduled predictor in which a state-preserving temporal lifting map is complemented by a context-dependent low-rank transition and a measurement-dependent input modulation. The transition and input matrices are bounded functions of a history context and the current lifted measurement, respectively. For fixed scheduling signals the predictor is linear in the lifted state and input, while recursive scheduling remains generally nonlinear. We analyze this distinction directly. Under common-history, work-domain, Lipschitz, and one-step residual conditions, we derive a finite-horizon open-loop prediction bound. For control, a nominal rollout supplies scheduling matrices that are frozen over the horizon, yielding an LTV prediction model and a convex quadratic program under polyhedral input constraints. We further derive a finite-horizon bound between the complete scheduled model and its frozen approximation, and state a conditional cost-suboptimality result relative to the global optimum of the complete problem. The results are deliberately limited to finite-horizon open-loop prediction and conditional optimization error; they do not assert closed-loop stability, recursive feasibility, real-time performance, or hardware-control guarantees.**

***Index Terms*—Koopman modeling, temporal lifting, scheduled prediction, low-rank modulation, LTV-MPC, model predictive control.**

I. Introduction

Learning a predictive model for a nonlinear controlled system is useful in a control setting when the model can be evaluated recursively and incorporated into a control calculation. Classical system identification provides a broad collection of state-space and input–output tools [4]–[6]. Koopman methods offer a complementary viewpoint: nonlinear state evolution is represented through linear evolution of observables, and finite-dimensional approximations can be used for prediction and control [1], [3], [8], [16], [29]. The finite-dimensional approximation, however, depends on the selected coordinates and on how operating-condition variation is represented.

When an instantaneous measurement is not a sufficient description of the relevant dynamics, delay coordinates and recurrent representations can provide additional temporal information [10]–[13], [51], [52]. A temporal lifting map can therefore encode a finite measurement history rather than a single sample. This addresses a representation problem but does not by itself solve a transition-model problem: all histories may still be propagated by one global lifted matrix. Context-dependent Koopman models and related temporal operator constructions instead allow the effective transition to vary with a recent trajectory or operating condition [14], [15], [37]. The present work combines these two ideas while retaining a structured, bounded low-rank modulation around a shared nominal transition.

Recursive prediction creates a second difficulty. The history used to train a one-step map is formed from measured samples, whereas deployment recursively updates the history with predicted measurements. The resulting context can differ from the context seen during one-step training. Thus, a small one-step residual does not by itself provide a multi-step guarantee. The analysis below makes the shared initial history and the recursive window update explicit, and treats the resulting delay coupling rather than replacing it with an instantaneous error recurrence.

The same scheduling dependence creates a control trade-off. If the scheduling variables are recomputed from the decision-dependent predicted trajectory, the complete finite-horizon problem is generally nonlinear and nonconvex. Freezing the scheduling matrices along a nominal rollout produces an LTV approximation and a convex quadratic program for polyhedral input constraints. This is a modeling approximation, not an assertion that the complete scheduled problem is linear. The distinction is central to the control analysis.

This construction is related to gain scheduling and quasi-linear parameter-varying control, where a linear model is indexed by an operating condition or scheduling signal [47]–[50]. It also differs from real-time iteration methods, which perform a single sequential-quadratic-programming update for a nonlinear optimal-control problem [43], [44]. Here the frozen matrices are generated from a history-aware nominal rollout and the result is analyzed as a finite-horizon LTV approximation. The paper therefore does not transfer the stability, convergence, or timing guarantees that may hold for particular gain-scheduled or RTI schemes.

The contributions are as follows.

1) We formulate a state-preserving, history-conditioned scheduled predictor with a low-rank bounded modulation of a nominal transition and a separate measurement-dependent input modulation. The two scheduling variables are separated as a modeling choice; no parameter-identifiability claim is made.
2) We derive a finite-horizon prediction-error recurrence that includes both the current lifted-state error and the

maximum error in the recent history window. Under a common initial history and a uniform one-step residual bound, this yields an explicit finite-horizon open-loop bound.

3) We formulate a frozen-LTV-MPC approximation using a nominal rollout, derive its prediction matrices and convex quadratic objective in measurement space, and bound the difference between complete and frozen trajectories for the same candidate input sequence. A conditional cost-suboptimality theorem follows when a uniform cost-approximation condition and common feasible set are available.

The paper has six sections. Section II introduces notation and the prediction/control tasks. Section III defines the scheduled predictor, training objective, recursive rollout, and prediction analysis. Section IV develops the frozen-LTV controller and its error results. Section V records the experimental protocol while intentionally leaving numerical results for a later, separately verified stage. Section VI summarizes the established scope and limitations.

II. Preliminaries and Problem Formulation

A. Coordinates, controlled Koopman representation, and notation

Consider a discrete-time controlled system

$$\mathbf{s}_{k+1} = f(\mathbf{s}_k, \mathbf{u}_k), \qquad \mathbf{x}_k = h(\mathbf{s}_k), \tag{1}$$

where $\mathbf{s}_k \in \mathcal{S}$ is a physical state, $\mathbf{u}_k \in \mathbb{R}^m$ is the input, and $\mathbf{x}_k \in \mathbb{R}^n$ is the available measurement. For a fixed input, the controlled Koopman operator acts on an observable φ according to

$$(\mathcal{K}_{\mathbf{u}}\varphi)(\mathbf{s}) = \varphi(f(\mathbf{s}, \mathbf{u})). \tag{2}$$

It is linear in the observable function, although the underlying state dynamics need not be linear. The finite-dimensional matrices used in this paper are a controlled prediction parameterization in selected lifted coordinates. They are not identified as an exact finite-dimensional truncation of the infinite-dimensional operator on the physical state.

For a vector a, let $\|a\|_2$ denote the Euclidean norm; for a matrix, $\|M\|_2$ denotes its spectral norm. For a positive semidefinite matrix M, write $\|a\|_M^2 = a^\top M a$. The symbol P denotes the history length, whereas P_x denotes a terminal cost matrix; these quantities are distinct.

B. Finite histories and prediction/control tasks

Let $P \geq 2$. At time k, define the measurement and past-input windows

$$\mathbf{X}_k = \begin{bmatrix} \mathbf{x}_{k-P+1} \\ \vdots \\ \mathbf{x}_k \end{bmatrix} \in \mathbb{R}^{Pn}, \qquad \mathbf{U}_k^{\mathrm{in}} = \begin{bmatrix} \mathbf{u}_{k-P+1} \\ \vdots \\ \mathbf{u}_{k-1} \end{bmatrix} \in \mathbb{R}^{(P-1)m}. \tag{3}$$

The current input $\mathbf{u}_k$ is not part of the context window; it enters the one-step update separately. Given $(\mathbf{X}_k, \mathbf{U}_k^{\mathrm{in}})$ and a future input sequence $\mathbf{u}_{k:k+H-1}$, the prediction task is to estimate $\mathbf{x}_{k+1:k+H}$. The control task is to choose a finite sequence $U = (\mathbf{u}_0, \ldots, \mathbf{u}_{N-1})$ that tracks a measurement reference while respecting the input set.

The lifting is state-preserving in the measured coordinates:

$$\mathbf{z}_k = g_\theta(\mathbf{X}_k) = \begin{bmatrix} \mathbf{x}_k \\ \psi_\theta(\mathbf{X}_k) \end{bmatrix} \in \mathbb{R}^d, \qquad d = n + n_\psi, \qquad \mathbf{x}_k = C_x \mathbf{z}_k, \tag{4}$$

where $C_x = [I_n \ 0]$. We distinguish $\mathbf{z}_k^{\mathrm{enc}} = g_\theta(\mathbf{X}_k)$, which uses measured data, from a recursively propagated $\mathbf{z}_k^{\mathrm{prop}}$, whose measured coordinates are model outputs.

The model and the control analysis use a common-history convention. All compared rollouts start from the same measured window and past-input window and from the same $\mathbf{z}_0 = g_\theta(\mathbf{X}_k)$. For a predicted index $j < 0$, a history slot reads the corresponding measured past sample rather than a fictitious negative-time latent state. After each step, the oldest measurement slot is replaced by $C_x \mathbf{z}_{i+1}$ and the oldest input slot by $\mathbf{u}_i$.

III. Scheduled Koopman Modeling

A. Model construction

The intended architecture is a patch-based temporal Transformer encoder [35], [36]. We impose the implementation constraint $p \mid P$: the history is partitioned into P/p contiguous patches, each patch is linearly embedded, augmented with positional encoding, processed by a Transformer encoder, and pooled before projection to the hidden lifted coordinates. The current measurement is concatenated to this projection as in (4). Thus, the theoretical map g_θ abstracts the architecture while the state-preserving readout and the recursive propagation interface remain explicit. The context map C_ϕ is a separate feed-forward map of the flattened measurement and past-input windows. These architectural details specify the intended temporal map; the bounds below use only the stated Lipschitz and work-domain conditions.

The history context is

$$\mathbf{c}_k = C_\phi(\mathbf{X}_k, \mathbf{U}_k^{\mathrm{in}}) \in \mathbb{R}^{d_c}, \qquad \boldsymbol{\chi}_k = C_x \mathbf{z}_k \in \mathbb{R}^n. \tag{5}$$

The scheduled predictor is

$$\boxed{\mathbf{z}_{k+1} = \mathbf{A}(\mathbf{c}_k)\mathbf{z}_k + \mathbf{B}(\boldsymbol{\chi}_k)\mathbf{u}_k.} \tag{6}$$

The map $\mathbf{A}$ is scheduled by history, while $\mathbf{B}$ is scheduled by the current measured component of the lifted state. This separation is a parameterization choice: it does not assert that the physical input effect is independent of history and does not, by itself, establish parameter identifiability.

We use bounded low-rank corrections around nominal matrices,

$$\mathbf{A}(\mathbf{c}) = \mathbf{A}_0 + \mathbf{U}_A \operatorname{diag}(\gamma_A(\mathbf{c}))\mathbf{V}_A^\top, \tag{7}$$

$$\mathbf{B}(\boldsymbol{\chi}) = \mathbf{B}_0 + \mathbf{U}_B \operatorname{diag}(\gamma_B(\boldsymbol{\chi}))\mathbf{V}_B^\top. \tag{8}$$

Here $\mathbf{A}_0 \in \mathbb{R}^{d\times d}$, $\mathbf{B}_0 \in \mathbb{R}^{d\times m}$, $\mathbf{U}_A, \mathbf{V}_A \in \mathbb{R}^{d\times r_A}$, $\mathbf{U}_B \in \mathbb{R}^{d\times r_B}$, and $\mathbf{V}_B \in \mathbb{R}^{m\times r_B}$. The transition coefficients have the form

$$\gamma_{A,j}(\mathbf{c}) = g_A \pi_{A,j}(\mathbf{c}) \tanh([W_A \mathbf{c} + b_A]_j), \quad \pi_A(\mathbf{c}) = \operatorname{softmax}(W_A^\pi \mathbf{c} + b_A^\pi) \tag{9}$$

and analogously for $\gamma_B(\boldsymbol{\chi})$, with g_B, W_B, and W_B^π. The columns of the four low-rank factors have norm at most one.

Proposition 1 (Uniform modulation bounds)**.** *For every* $\mathbf{c} \in \mathbb{R}^{d_c}$ *and* $\boldsymbol{\chi} \in \mathbb{R}^n$*,*

$$\|\mathbf{A}(\mathbf{c}) - \mathbf{A}_0\|_2 \le g_A, \qquad \|\mathbf{B}(\boldsymbol{\chi}) - \mathbf{B}_0\|_2 \le g_B. \tag{10}$$

Proof. Writing the correction as a sum of rank-one matrices and applying the triangle inequality gives $\|\mathbf{A}(\mathbf{c}) - \mathbf{A}_0\|_2 \le \sum_j |\gamma_{A,j}(\mathbf{c})|$. Softmax weights are nonnegative and sum to one, while $|\tanh(\cdot)| < 1$; hence $|\gamma_{A,j}(\mathbf{c})| \le g_A \pi_{A,j}(\mathbf{c})$. The result follows. The argument for $\mathbf{B}$ is identical. □

This is a boundedness result, not a stability result. In particular, it does not imply $\rho(\mathbf{A}(\mathbf{c})) < 1$, contraction of time-varying matrix products, closed-loop stability, or recursive feasibility.

The schedule maps are also Lipschitz under finite modulation-layer weights. Define

$$L_A = g_A \left(\|W_A\|_2 + \frac{\sqrt{r_A}}{2} \|W_A^\pi\|_2 \right), \quad L_B = g_B \left(\|W_B\|_2 + \frac{\sqrt{r_B}}{2} \|W_B^\pi\|_2 \right). \tag{11}$$

Then

$$\|\mathbf{A}(\mathbf{c}) - \mathbf{A}(\mathbf{c}')\|_2 \le L_A \|\mathbf{c} - \mathbf{c}'\|_2, \quad \|\mathbf{B}(\boldsymbol{\chi}) - \mathbf{B}(\boldsymbol{\chi}')\|_2 \le L_B \|\boldsymbol{\chi} - \boldsymbol{\chi}'\|_2. \tag{12}$$

The factor $1/2$ is the global Euclidean Lipschitz constant of softmax for $r \ge 2$; the full constants in (11) may still be conservative because they combine norm inequalities and triangle inequalities.

B. Training and recursive prediction

The training objective keeps the measurement and lifted coordinates coupled. For a training transition or rollout, we use the joint form

$$\mathcal{L} = \mathcal{L}_x + w_z \mathcal{L}_z, \qquad w_z \ge 0, \tag{13}$$

where $\mathcal{L}_x$ measures the measured prediction error and $\mathcal{L}_z$ measures the corresponding lifted-state error. The two terms serve different purposes: the first controls the observable output used by prediction and control, while the second regularizes the selected lifted coordinates. This paper does not treat adding the lifted-state term as an ablation or claim that it is empirically superior.

At deployment, the model is recursively propagated. Starting from $\mathbf{z}_0 = g_\theta(\mathbf{X}_k)$, construct the predicted history from the model's own outputs and use

$$\mathbf{z}_{i+1} = \mathbf{A}(\mathbf{c}_i^{\mathrm{prop}})\mathbf{z}_i + \mathbf{B}(C_x \mathbf{z}_i)\mathbf{u}_i, \tag{14}$$

where $\mathbf{c}_i^{\mathrm{prop}}$ is computed from the recursively shifted measurement and input windows. No future measured state is used in this recursion.

C. Finite-horizon prediction analysis

We state the prediction result in a form that makes its scope explicit. Let $\mathbf{z}_{k+i}^{\mathrm{enc}}$ be the encoder sequence formed from the true measurement windows and let $\hat{\mathbf{z}}_i$ be the propagated sequence under the same input sequence. Define

$$\mathbf{e}_i = \mathbf{z}_{k+i}^{\mathrm{enc}} - \hat{\mathbf{z}}_i, \qquad \mathbf{e}_0 = 0. \tag{15}$$

Assume that all relevant encoded and propagated trajectories remain in a common finite work domain, that C_ϕ is Lipschitz with constant L_ϕ, and that the one-step latent residual

$$\mathbf{r}_k = \mathbf{z}_{k+1}^{\mathrm{enc}} - \mathbf{A}(\mathbf{c}_k)\mathbf{z}_k^{\mathrm{enc}} - \mathbf{B}(\boldsymbol{\chi}_k)\mathbf{u}_k \tag{16}$$

obeys $\|\mathbf{r}_k\|_2 \le \delta$ on the considered domain and horizon. Let

$$M_A = \|\mathbf{A}_0\|_2 + g_A, \qquad L = M_A + L_B M_U + L_A L_\phi \sqrt{P} M_Z, \tag{17}$$

where M_Z and M_U are common state and input bounds.

For traceability, the assumptions used below are stated formally. They are local assumptions on a fixed finite horizon, not global dynamical guarantees.

Assumption 1 (A1: common finite work domain)**.** *For the fixed initial history, parameters, and horizon, there exist compact sets* $\mathcal{Z}$*,*$\mathcal{C}$*, and* $\mathcal{X}$*, and a nonempty compact convex input set* $\mathcal{U}$*, such that every encoded, propagated, nominal, complete, and frozen trajectory used in the relevant comparison remains in* $\mathcal{Z}$*, its scheduling signal remains in* $\mathcal{C}$*, and its measured window remains in the common domain of* C_ϕ*. Let* $M_Z = \sup_{z \in \mathcal{Z}} \|z\|_2$ *and* $M_U = \sup_{u \in \mathcal{U}} \|u\|_2$*.*

Assumption 2 (A2: lifting regularity)**.** g_θ *is Lipschitz on a named domain* $\mathcal{D}_g$ *containing all encoder windows considered in the finite horizon:* $\|g_\theta(W) - g_\theta(W')\|_2 \le L_g \|W - W'\|_2$ *for* $W, W' \in \mathcal{D}_g$*. A2 is used only if the lifting residual is separated from* $\mathbf{r}_k$*; it is not used in Theorem 1.*

Assumption 3 (A3: common-window context regularity)**.** *On the common window domain in A1,* C_ϕ *is Lipschitz with one constant* L_ϕ *for every pair of windows:*

$$\|C_\phi(W, V) - C_\phi(W', V')\|_2 \le L_\phi \|(W, V) - (W', V')\|_2. \tag{18}$$

Assumption 4 (A4: uniform modulation weights)**.** *When a family of trained models is considered, there are finite constants* M_{W_A}*,* $M_{W_A^\pi}$*,* M_{W_B}*, and* $M_{W_B^\pi}$ *such that*

$$\|W_A\|_2 \le M_{W_A}, \quad \|W_A^\pi\|_2 \le M_{W_A^\pi}, \quad \|W_B\|_2 \le M_{W_B}, \quad \|W_B^\pi\|_2 \le M$$

(19)

For a fixed trained model, A4 is only the finite-constant condition used in (11)*.*

Assumption 5 (A5: uniform one-step residual)**.** *For all encoded transitions* $k + i$ *with* $i = 0, \ldots, H - 1$ *in the common work domain, the residual in* (16) *satisfies* $\|\mathbf{r}_{k+i}\|_2 \le \delta$ *for one common* $\delta \ge 0$*.*

Theorem 1 uses A1, A3, and A5 together with the structural bounds; it does not use A2. The freezing bound uses A1 and A3. The conditional cost theorem additionally requires one common feasible set and a uniform cost approximation over that set.

Proposition 2 (One-step prediction recurrence)**.** *For every* $0 \le i \le H-1$*,*

$$\|\mathbf{e}_{i+1}\|_2 \le \delta + (M_A + L_B M_U)\,\|\mathbf{e}_i\|_2 + L_A L_\phi \sqrt{P} M_Z \max_{i-P+1\le j\le i} \|\mathbf{e}_j\|_2 , \tag{20}$$

where negative-index history errors are defined as zero because the initial measured history is shared.

Proof. Subtract the propagated update from the encoder update and add and subtract the encoder-scheduled terms:

$$\mathbf{e}_{i+1} = \mathbf{r}_{k+i} + \left[\mathbf{A}(\mathbf{c}_{k+i}^{\mathrm{enc}})\mathbf{z}_{k+i}^{\mathrm{enc}} - \mathbf{A}(\mathbf{c}_i^{\mathrm{prop}})\hat{\mathbf{z}}_i\right] \tag{21}$$

$$+ \left[\mathbf{B}(\boldsymbol{\chi}_{k+i}^{\mathrm{enc}}) - \mathbf{B}(\boldsymbol{\chi}_i^{\mathrm{prop}})\right]\mathbf{u}_{k+i}. \tag{22}$$

The first term is bounded by A5. Splitting the second term into state and schedule differences gives $M_A \|\mathbf{e}_i\|_2$ plus $L_A M_Z$ times the context difference. The input portions of the two windows are identical, while the measurement portions differ by at most $\sqrt{P}\max_{i-P+1\le j\le i}\|\mathbf{e}_j\|_2$; A3 therefore bounds the context difference by L_ϕ times this quantity. Finally, A3 is not needed for the input branch: (12) and $\|\mathbf{u}_{k+i}\|_2 \le M_U$ give $L_B M_U \|\mathbf{e}_i\|_2$. Combining the three bounds proves (20). □

Theorem 1 (Finite-horizon open-loop prediction bound)**.** *Under the conditions above, for every* $0 \le i \le H$*,*

$$\|\mathbf{e}_i\|_2 \le W_i \le \delta \sum_{t=0}^{i-1} L^{i-1-t}, \qquad W_i = \max_{0\le j\le i} \|\mathbf{e}_j\|_2 . \tag{23}$$

Thus, for $L \ne 1$*,* $\|\mathbf{e}_H\|_2 \le \delta(L^H - 1)/(L-1)$*, and for* $L = 1$*,* $\|\mathbf{e}_H\|_2 \le H\delta$*.*

Proof. The residual, state-propagation, schedule-difference, and measurement-dependent input terms give (20). Since the recent-window maximum is bounded by W_i, one obtains $\|\mathbf{e}_{i+1}\|_2 \le \delta + LW_i$. Let $S_0 = 0$ and $S_{i+1} = \delta + LS_i$. The sequence $S_i = \delta\sum_{t=0}^{i-1} L^{i-1-t}$ is nondecreasing for $L \ge 0$, since $S_{i+1} - S_i = \delta L^i$. Induction in $W_{i+1} = \max(W_i, \|\mathbf{e}_{i+1}\|_2)$ yields $W_i \le S_i$. □

The result is an open-loop, finite-horizon statement. It does not require $L < 1$ and does not imply an infinite-horizon bound. The uniform residual and work-domain conditions are assumptions requiring separate certification; a finite sample maximum is not automatically a uniform bound over the work domain.

IV. Frozen-LTV Model Predictive Control

A. Complete scheduled problem and freezing construction

At a control instant, let $\mathbf{z}_0 = \mathbf{z}_k^{\mathrm{enc}}$ and let $U = (\mathbf{u}_0, \dots, \mathbf{u}_{N-1})$. Assume the input set $\mathcal{U} \subset \mathbb{R}^m$ is nonempty, compact, and convex; in the standard QP case it is a box or another polytope. For the input-only formulation, define the common feasible set $\mathcal{U}_{\mathrm{feas}} = \mathcal{U}^N$. The complete scheduled model uses

$$\mathbf{z}_{i+1} = \mathbf{A}(\mathbf{c}_i)\mathbf{z}_i + \mathbf{B}(C_x\mathbf{z}_i)\mathbf{u}_i, \tag{24}$$

with history slots shifted according to Section II. Given measurement references $\mathbf{x}_i^{\mathrm{ref}}$, input references $\mathbf{u}_i^{\mathrm{ref}}$, previous input $\mathbf{u}_{-1}$, and positive semidefinite weights Q_x, P_x, R, S, define

$$J_{\mathrm{full}}(U) = \sum_{i=1}^{N-1} \left\|C_x\mathbf{z}_i - \mathbf{x}_i^{\mathrm{ref}}\right\|_{Q_x}^2 + \left\|C_x\mathbf{z}_N - \mathbf{x}_N^{\mathrm{ref}}\right\|_{P_x}^2 \tag{25}$$

$$+ \sum_{i=0}^{N-1} \left\|\mathbf{u}_i - \mathbf{u}_i^{\mathrm{ref}}\right\|_R^2 + \sum_{i=0}^{N-1} \|\mathbf{u}_i - \mathbf{u}_{i-1}\|_S^2 . \tag{26}$$

The state tracking cost is deliberately expressed in measurement space. No reference is assigned to hidden lifted coordinates. The complete problem is generally nonlinear and nonconvex because the scheduling variables depend on the decision-dependent trajectory.

Choose a feasible nominal input sequence $\bar{U} = (\bar{\mathbf{u}}_0, \dots, \bar{\mathbf{u}}_{N-1})$ and initialize the nominal rollout with the same state and pre-horizon history, $\bar{\mathbf{z}}_0 = \mathbf{z}_0$, $\bar{X}_0 = \mathbf{X}_k$, and $\bar{U}_0^{\mathrm{in}} = \mathbf{U}_k^{\mathrm{in}}$. At each step set $\bar{\boldsymbol{\chi}}_i = C_x\bar{\mathbf{z}}_i$ and $\bar{\mathbf{c}}_i = C_\phi(\bar{X}_i, \bar{U}_i^{\mathrm{in}})$, then roll out the complete model once:

$$\bar{\mathbf{z}}_{i+1} = \mathbf{A}(\bar{\mathbf{c}}_i)\bar{\mathbf{z}}_i + \mathbf{B}(\bar{\boldsymbol{\chi}}_i)\bar{\mathbf{u}}_i. \tag{27}$$

Set

$$\bar{\mathbf{A}}_i = \mathbf{A}(\bar{\mathbf{c}}_i), \qquad \bar{\mathbf{B}}_i = \mathbf{B}(\bar{\boldsymbol{\chi}}_i), \tag{28}$$

and keep these matrices fixed while optimizing the candidate sequence U. The frozen trajectory is therefore

$$\mathbf{z}_{i+1}^{\mathrm{fr}} = \bar{\mathbf{A}}_i\mathbf{z}_i^{\mathrm{fr}} + \bar{\mathbf{B}}_i\mathbf{u}_i. \tag{29}$$

The nominal rollout, complete trajectory, and frozen trajectory are distinct objects: they use different input sequences and/or different scheduling rules.

For the uniform freezing and cost statements below, fix $\bar{U} \in \mathcal{U}_{\mathrm{feas}}$ and require A1 and A3 to hold simultaneously for every candidate $U \in \mathcal{U}_{\mathrm{feas}}$, with the same $\mathcal{Z}$, $\mathcal{C}_\phi$ window domain, M_Z, M_U, and L_ϕ. This is a hypothesis on the whole feasible input set, not an additional state constraint in the QP.

B. Prediction matrices and convex quadratic program

Stack the frozen states and inputs as

$$\mathbf{Z} = [(\mathbf{z}_1^{\mathrm{fr}})^\top, \dots, (\mathbf{z}_N^{\mathrm{fr}})^\top]^\top, \qquad \mathbf{U} = [\mathbf{u}_0^\top, \dots, \mathbf{u}_{N-1}^\top]^\top. \tag{30}$$

The frozen model gives

$$\mathbf{Z} = \mathbf{F} + \mathbf{G}\mathbf{U}, \tag{31}$$

where the block rows of $\mathbf{F}$ and $\mathbf{G}$ are denoted by F_i and $G_{i,t}$, respectively:

$$F_i = \bar{\mathbf{A}}_{i-1}\cdots\bar{\mathbf{A}}_0\mathbf{z}_0, \qquad G_{i,t} = \begin{cases} \bar{\mathbf{A}}_{i-1}\cdots\bar{\mathbf{A}}_{t+1}\bar{\mathbf{B}}_t, & t < i, \\ 0, & t \ge i. \end{cases} \tag{32}$$

The products are time ordered; no commutativity of the matrices is assumed. Let $\bar{C}_x = I_N \otimes C_x$, $Y^{\mathrm{ref}} = [(\mathbf{x}_1^{\mathrm{ref}})^\top, \dots, (\mathbf{x}_N^{\mathrm{ref}})^\top]^\top$, and $\bar{Q}_x = \mathrm{blkdiag}(Q_x, \dots, Q_x, P_x)$. Define $\bar{R}$ and $\bar{S}$ as the corresponding block-diagonal input matrices, and let $\hat{D}$ satisfy $(\hat{D}\mathbf{U})_0 = \mathbf{u}_0$ and $(\hat{D}\mathbf{U})_i = \mathbf{u}_i - \mathbf{u}_{i-1}$

for $i \geq 1$. With the input-offset vector $\hat{\mathbf{c}} = [-\mathbf{u}_{-1}^\top, 0, \ldots, 0]^\top$, the frozen cost is

$$J_{\mathrm{fr}}(U;\bar{U}) = \left\|\bar{C}_x\mathbf{Z} - Y^{\mathrm{ref}}\right\|_{\bar{Q}_x}^2 + \left\|\mathbf{U} - \mathbf{U}^{\mathrm{ref}}\right\|_{\bar{R}}^2 + \left\|\hat{D}\mathbf{U} + \hat{\mathbf{c}}\right\|_{\bar{S}}^2. \tag{33}$$

Substitution of (31) gives

$$J_{\mathrm{fr}}(U;\bar{U}) = \frac{1}{2}\mathbf{U}^\top H\mathbf{U} + 2f^\top\mathbf{U} + \mathrm{const}, \tag{34}$$

where

$$H = 2\left(\mathbf{G}^\top\bar{C}_x^\top\bar{Q}_x\bar{C}_x\mathbf{G} + \bar{R} + \hat{D}^\top\bar{S}\hat{D}\right), \tag{35}$$

$$f = \mathbf{G}^\top\bar{C}_x^\top\bar{Q}_x(\bar{C}_x\mathbf{F} - Y^{\mathrm{ref}}) - \bar{R}\mathbf{U}^{\mathrm{ref}} + \hat{D}^\top\bar{S}\hat{\mathbf{c}}. \tag{36}$$

Every term in H is positive semidefinite, so the objective is convex. With a box or other polyhedral input set, the frozen problem is a standard convex QP. Strict convexity and uniqueness require an additional positive-definiteness condition, for example $\bar{R} \succ 0$. The use of a numerical QP solver does not, by itself, certify exact global optimality of its returned iterate.

C. Freezing error and conditional suboptimality

Fix a candidate U and nominal $\bar{U}$. Let $\mathbf{z}_i^{\mathrm{full}}$ be the complete scheduled trajectory, $\mathbf{z}_i^{\mathrm{fr}}$ the frozen trajectory, and $\bar{\mathbf{z}}_i$ the nominal trajectory. For $i = 0, \ldots, N-1$, let U_i^{in} and $\bar{U}_i^{\mathrm{in}}$ denote the length-$(P-1)$ past-input windows assembled from the common pre-horizon inputs and from $\mathbf{u}_0, \ldots, \mathbf{u}_{i-1}$ or $\bar{\mathbf{u}}_0, \ldots, \bar{\mathbf{u}}_{i-1}$, respectively. Define $\mathbf{e}_i = \mathbf{z}_i^{\mathrm{full}} - \mathbf{z}_i^{\mathrm{fr}}$ and $\mathbf{g}_i = \mathbf{z}_i^{\mathrm{fr}} - \bar{\mathbf{z}}_i$. Let

$$\delta_u = \max_{0\leq i<N} \|\mathbf{u}_i - \bar{\mathbf{u}}_i\|_2, \qquad \Delta_u = \max_{0\leq i<N} \left\|U_i^{\mathrm{in}} - \bar{U}_i^{\mathrm{in}}\right\|_2. \tag{37}$$

Then $\Delta_u \leq \sqrt{P-1}\delta_u$ because the past input windows share their initial entries and differ only in shifted future inputs. With $M_B = \|\mathbf{B}_0\|_2 + g_B$, the nominal drift satisfies

$$\|\mathbf{g}_i\|_2 \leq G_i := M_B\delta_u \sum_{t=0}^{i-1} M_A^{i-1-t}. \tag{38}$$

To make the bound uniform over the complete feasible set, define the fixed-nominal suprema

$$\bar{\delta}_u := \sup_{U\in\mathcal{U}_{\mathrm{feas}}} \max_{0\leq i<N} \|\mathbf{u}_i - \bar{\mathbf{u}}_i\|_2, \qquad \bar{\Delta}_u := \sup_{U\in\mathcal{U}_{\mathrm{feas}}} \max_{0\leq i<N} \left\|U_i^{\mathrm{in}} - \bar{U}_i^{\mathrm{in}}\right\|_2. \tag{39}$$

Compactness of $\mathcal{U}_{\mathrm{feas}}$ makes these quantities finite, and $\bar{\Delta}_u \leq \sqrt{P-1}\bar{\delta}_u$. Set

$$\bar{G}_N := M_B\bar{\delta}_u \sum_{t=0}^{N-1} M_A^t, \qquad \bar{\delta}_{\mathrm{frz}} := L_A L_\phi M_Z\bar{\Delta}_u + \left(L_A L_\phi\sqrt{P}M_Z + L_B M_U\right)\bar{G}_N, \tag{40}$$

and, for $0 \leq i \leq N$,

$$\bar{K}_i := \bar{\delta}_{\mathrm{frz}} \sum_{t=0}^{i-1} L^{i-1-t}. \tag{41}$$

For every $U \in \mathcal{U}_{\mathrm{feas}}$, the candidate-specific quantities satisfy $\delta_u \leq \bar{\delta}_u$, $\Delta_u \leq \bar{\Delta}_u$, $G_N \leq \bar{G}_N$, and hence $\|\mathbf{e}_i\|_2 \leq \bar{K}_i$.

For a common work domain, define

$$\delta_{\mathrm{frz}} = L_A L_\phi M_Z\Delta_u + \left(L_A L_\phi\sqrt{P}M_Z + L_B M_U\right) G_N. \tag{42}$$

With L from (17), the freezing error obeys

$$\|\mathbf{e}_i\|_2 \leq K_i := \delta_{\mathrm{frz}} \sum_{t=0}^{i-1} L^{i-1-t}, \qquad 0 \leq i \leq N. \tag{43}$$

This is a same-input, finite-horizon open-loop comparison. In particular, it is not a comparison of two optimization algorithms.

To express the cost difference uniformly, let $M_Y = \|C_x\|_2 M_Z$, $M_{\mathrm{ref}} = \max_{1\leq i\leq N} \left\|\mathbf{x}_i^{\mathrm{ref}}\right\|_2$, and $\bar{K}_i^x = \|C_x\|_2 \bar{K}_i$. A uniform sufficient bound is

$$\varepsilon_J = 2(M_Y + M_{\mathrm{ref}}) \left[\|Q_x\|_2 \sum_{i=1}^{N-1} \bar{K}_i^x + \|P_x\|_2 \bar{K}_N^x\right] \tag{44}$$

$$+ \|Q_x\|_2 \sum_{i=1}^{N-1} (\bar{K}_i^x)^2 + \|P_x\|_2 (\bar{K}_N^x)^2. \tag{45}$$

It follows from the identity for the difference of two squared weighted norms and from the fact that the input and input-increment terms in (26) and (33) are identical.

More explicitly, let $y_i^{\mathrm{full}} = C_x\mathbf{z}_i^{\mathrm{full}}$, $y_i^{\mathrm{fr}} = C_x\mathbf{z}_i^{\mathrm{fr}}$, and $d_i = y_i^{\mathrm{full}} - y_i^{\mathrm{fr}}$. The uniform freezing bound gives $\|d_i\|_2 \leq \bar{K}_i^x$. With $M_Y = \|C_x\|_2 M_Z$ and $M_{\mathrm{ref}} = \max_{1\leq i\leq N} \left\|\mathbf{x}_i^{\mathrm{ref}}\right\|_2$, A1 gives $\left\|y_i^{\mathrm{full}}\right\|_2 \leq M_Y$ and hence $\left\|y_i^{\mathrm{fr}}\right\|_2 \leq M_Y + \bar{K}_i^x$. Therefore, for $i < N$,

$$\left|\left\|y_i^{\mathrm{full}} - \mathbf{x}_i^{\mathrm{ref}}\right\|_{Q_x}^2 - \left\|y_i^{\mathrm{fr}} - \mathbf{x}_i^{\mathrm{ref}}\right\|_{Q_x}^2\right| \leq \|Q_x\|_2 \bar{K}_i^x(2M_Y + \bar{K}_i^x + 2M_{\mathrm{ref}}), \tag{46}$$

and the same identity with P_x applies at $i = N$. Summing these bounds and using the common input terms yields (45). Because (39) takes the supremum over the entire feasible set and A1/A3 use the same domain constants there, this calculation proves the uniform candidate bound required by (CA), subject to the stated work-domain assumptions.

Assumption 6 (Uniform cost approximation)**.** *For the fixed nominal sequence, the complete and frozen costs obey*

$$\sup_{U\in\mathcal{U}_{\mathrm{feas}}} |J_{\mathrm{full}}(U) - J_{\mathrm{fr}}(U;\bar{U})| \leq \varepsilon_J. \tag{47}$$

Theorem 2 (Conditional cost suboptimality)**.** *Suppose the complete and frozen problems have the same nonempty feasible input set, and let U^{fr} and $U^\star$ be global minimizers of the frozen and complete costs, respectively. Under Assumption 6,*

$$0 \leq J_{\mathrm{full}}(U^{\mathrm{fr}}) - J_{\mathrm{full}}(U^\star) \leq 2\varepsilon_J. \tag{48}$$

Proof. The lower bound follows from global optimality of $U^\star$. For the upper bound,

$$\begin{aligned} J_{\mathrm{full}}(U^{\mathrm{fr}}) &\leq J_{\mathrm{fr}}(U^{\mathrm{fr}};\bar{U}) + \varepsilon_J \\ &\leq J_{\mathrm{fr}}(U^\star;\bar{U}) + \varepsilon_J \\ &\leq J_{\mathrm{full}}(U^\star) + 2\varepsilon_J. \end{aligned}$$

□

The theorem is conditional. It does not certify Assumption 6 from finite samples, does not cover a numerically approximate QP solution without an additional solver-error certificate, and does not imply closed-loop stability, recursive feasibility, real-time performance, or hardware-control performance.

V. Experiments

This section is intentionally a manuscript placeholder. Experimental results, figures, and numerical claims are omitted in the present draft and must be inserted only after the corresponding data, code version, and evidence have been independently checked. No statement in Sections I–IV relies on an unreported experimental result.

A. Data and protocol

The final version will specify the data source, sampling interval, state and input coordinates, history length, train/validation/test split, normalization, input constraints, and random seeds. The protocol must distinguish measurement coordinates from lifted coordinates and must report prediction horizon in both steps and physical time when the sampling interval is relevant.

B. Modeling evaluation

The final version will report one-step and multi-step measurement-space prediction errors, together with the joint training objective in (13). Any comparison among architectures or loss choices requires a fixed data split, fixed evaluation horizons, and an explicitly stated scope. The present theory does not claim that the lifted-state term improves empirical performance.

C. Control evaluation

The final version will describe the nominal-sequence policy, horizon, weights, input constraints, solver status, and whether each result is an offline simulation or a physical experiment. Offline closed-loop simulation must not be described as hardware validation. Timing measurements, if later reported, must be measured independently rather than inferred from the convex-QP formulation.

VI. Conclusion

This paper formulated a history-conditioned scheduled predictor with a state-preserving temporal lifting, a bounded low-rank history-conditioned transition, and a measurement-dependent input modulation. The scheduled form preserves linear propagation when the scheduling signals are fixed, while its recursive use remains generally nonlinear. The finite-horizon prediction result makes the history-window coupling explicit and gives a conditional open-loop bound under common-history, work-domain, Lipschitz, and residual conditions.

For control, freezing the scheduling matrices along a nominal rollout produces an LTV prediction model and a convex QP with measurement-space tracking costs. The freezing analysis separates nominal drift from the complete-versus-frozen trajectory error. Under a uniform cost approximation and a common feasible set, the frozen global optimizer satisfies a conditional $2\varepsilon_J$ cost bound relative to the global optimizer of the complete scheduled problem.

The scope is intentionally limited. The results do not establish stability, recursive feasibility, real-time deadlines, exact optimality of an approximate numerical solver, or physical hardware-control guarantees. The uniform work-domain and residual conditions require separate certification for any concrete system, and the experimental section remains to be completed from independently verified results.

Appendix A
Proof Details for the Scheduling Bounds

For completeness, the softmax Jacobian is $J(\pi) = \operatorname{diag}(\pi) - \pi\pi^\top$. It is positive semidefinite and, by the variance bound, satisfies $\|J(\pi)\|_2 \le 1/2$. Integrating the Jacobian along the line segment between two logits gives $\|\operatorname{softmax}(a) - \operatorname{softmax}(b)\|_2 \le \frac{1}{2}\|a-b\|_2$. Combining this inequality with the $\tanh$ Lipschitz property, $\|\cdot\|_1 \le \sqrt{r}\,\|\cdot\|_2$, and the rank-one factor normalization gives (12).

Appendix B
Proof Details for the Frozen Error Bound

The three trajectories in Section IV satisfy

$$\mathbf{e}_{i+1} = \bar{\mathbf{A}}_i\mathbf{e}_i + \big(\mathbf{A}(\mathbf{c}_i^{\mathrm{full}}) - \bar{\mathbf{A}}_i\big)\mathbf{z}_i^{\mathrm{full}} + \big(\mathbf{B}(\boldsymbol{\chi}_i^{\mathrm{full}}) - \bar{\mathbf{B}}_i\big)\mathbf{u}_i. \quad (49)$$

The history-window difference in the first matrix term is bounded by $L_\phi(\sqrt{P}(W_i + G_N) + \Delta_u)$. The measurement difference in the second matrix term is bounded by $L_B(W_i + G_N)$. Using the common state and input bounds gives

$$\|\mathbf{e}_{i+1}\|_2 \le \delta_{\mathrm{frz}} + LW_i. \quad (50)$$

The same nondecreasing comparison sequence used in the proof of Theorem 1 then gives (43).

Appendix C
Proof Details for the Quadratic Program

Let $\mathbf{G}_x = \bar{C}_x\mathbf{G}$ and $\mathbf{W} = \bar{C}_x\mathbf{F} - Y^{\mathrm{ref}}$. The output term expands as

$$(\mathbf{W}+\mathbf{G}_x\mathbf{U})^\top\bar{Q}_x(\mathbf{W}+\mathbf{G}_x\mathbf{U}) = \mathbf{U}^\top\mathbf{G}_x^\top\bar{Q}_x\mathbf{G}_x\mathbf{U} + 2(\mathbf{G}_x^\top\bar{Q}_x\mathbf{W})^\top\mathbf{U} + \mathrm{con} \quad (51)$$

The input and input-increment terms expand analogously. Their sum yields (36). Positive semidefiniteness follows because each quadratic coefficient has the form $T^\top MT$ with $M \succeq 0$.

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